ASSIGNMENT
IQR

DATASET:

Placement.csv

OUTPUT TABLE:

[24]:
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descriptive
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[24]:

	sl_no	ssc_p	hsc_p	degree_p	etest_p	mba_p	salary
Q1:25%	54.5	60.6	60.9	61.0	60.0	57.945	240000.0
Q3:75%	161.5	75.7	73.0	72.0	83.5	66.255	300000.0
IQR	107.0	15.1	12.1	11.0	23.5	8.31	60000.0
1.5rule	160.5	22.65	18.15	16.5	35.25	12.465	90000.0
Lesser	-106.0	37.95	42.75	44.5	24.75	45.48	150000.0
Greater	322.0	98.35	91.15	88.5	118.75	78.72	390000.0
Min	1	40.89	37.0	50.0	50.0	51.21	200000.0
Max	215	89.4	97.7	91.0	98.0	77.89	940000.0

OBSERVATION:

- From the table, we get the percentile of the particular students in their SSC, HSC, degree, etest, MBA, and salary

- SSC Percentile:
    - Q1 = 25%, where the value is 60.6
    - Q3 = 75%, where the value is 75.7
    - IQR = 15.1
    - 1.5rule = 22.65
    - Lesser = 37.95
    - Greater = 98.35
    - Min = 40.89
    - Max = 89.4

- HSC Percentile:
    - Q1 = 25%, where the value is 60.9
    - Q3 = 75%, where the value is 73.0
    - IQR = 12.1
    - 1.5rule = 18.15
    - Lesser = 42.75
    - Greater = 91.15
    - Min = 37
    - Max = 97.7

- Degree Percentile:
    - Q1 = 25%, where the value is 61.0
    - Q3 = 75%, where the value is 72.0
    - IQR = 11
    - 1.5rule = 16.5
    - Lesser = 44.5
    - Greater = 88.5
    - Min = 50
    - Max = 91

- Etest Percentile:
    - Q1 = 25%, where the value is 60.0
    - Q3 = 75%, where the value is 83.5
    - IQR = 23.5
    - 1.5rule = 35.25
    - Lesser = 24.75
    - Greater = 118.75
    - Min = 50.0
    - Max = 98.0

- MBA Percentile:
    - Q1 = 25%, where the value is 57.945
    - Q3 = 75%, where the value is 66.255
    - IQR = 8.31
    - 1.5rule = 12.465
    - Lesser = 45.48
    - Greater = 78.72
    - Min = 51.21
    - Max = 77.89

- Salary Percentile:
    - Q1 = 25%, where the value is 240000
    - Q3 = 75%, where the value is 300000
    - IQR = 60000
    - 1.5rule = 90000
    - Lesser = 150000
    - Greater = 390000
    - Min = 200000
    - Max = 940000

Interquartile Range(IQR)

a. The interquartile range. Compare the two interquartile ranges.

b. Any outliers in either set.

The five number summary for the day and night classes is

	Minimum	Q_1	Median	Q_3	Maximum
Day	32	56	74.5	82.5	99
Night	25.5	78	81	89	98

SOLUTION:

DAY:

IQR:

IQR = Q3-Q1

82.5 - 56

IQR = 26.5

LESSER OUTLIERS:

Q1 - 1.5 * IQR

56 - 1.5 * 26.5

LO = 29.5

<u>GREATER OUTLIERS:</u>

Q3 + 1.5 * IQR

82.5 + 1.5 * 26.5

GO = 122.25

Since the min value is 32, which is not less than “LO = 29.5”, and the max value is 99, which is not greater than “GO = 122.25”.

So the Day has no outliers.

<u>NIGHT:</u>

<u>IQR:</u>

Q3 - Q1

89 - 78

IQR = 11

<u>LESSER OUTLIERS:</u>

Q1 - 1.5 * IQR

78 - 1.5 * 11

LO = 61.5

<u>GREATER OUTLIERS:</u>

Q3 + 1.5 * IQR

89 + 1.5 * 11

GO = 105.5

Since the min value is 25.5, which is less than “LO = 61.5”, and the max value is 98, which is not greater than “GO = 105.5”.

So the Night has a Lesser Outlayer.